

DR.-ING. MARC NEU

Karlsruhe, Germany • +49 1590 1303147 • mail@marcneu.com • marcneu.github.io • ORCID 0000-0002-4564-8009

PROFESSIONAL SUMMARY

Hardware engineer specializing in real-time systems on FPGAs and heterogeneous SoCs, from deep learning accelerators for graph neural networks to high-throughput data acquisition. Currently leading the development of an MLIR-based compiler that lowers quantized ONNX models to HLS and AMD AI Engine targets. Designed the first dynamic point cloud network trigger in operation at a particle physics experiment, running at microsecond latency. Experienced in taking research prototypes into production, covering algorithm mapping, RTL implementation, verification, and on-site commissioning.

PROFESSIONAL EXPERIENCE

Postdoctoral Visiting Researcher

Oct – Dec 2026

High Energy Accelerator Research Organization (KEK), Tsukuba, Japan

- **Upgrade and commissioning:** Upgrading the GNN-ETM, an FPGA-based graph neural network trigger that selects collision events in real time at the Belle II experiment, and validating the upgrade on recorded data.

Hardware Research Engineer (Doctoral Researcher)

2022 – 2026

ITIV, Karlsruhe Institute of Technology (KIT), Karlsruhe

- **Dataflow compiler:** Leading the ongoing development of DeePloy, an MLIR-based compiler written in C++ that lowers quantized ONNX models to Vitis HLS C++ and AMD AI Engine kernels through custom dialects and quantization-aware passes.
- **Machine learning hardware trigger:** Designed and commissioned the GNN-ETM, an FPGA-based graph neural network trigger for the Belle II calorimeter. In operation since 2025, sustaining 8 million frames per second at an end-to-end latency of 1.067 μ s.
- **Acceleration on heterogeneous SoCs:** Mapped a dynamic point cloud network onto AMD Versal VCK190 by partitioning it across FPGA and AI Engines, fusing operators, and optimizing kernels. Achieved 7.02 $\times$ the throughput of an NVIDIA L40S GPU (IEEE FCCM 2026 Computing Challenge).
- **Joint communication and sensing DAQ:** Built a distributed massive MIMO OFDM data acquisition platform with 32 RFSocs, supporting up to 256 channels at 1 GHz baseband bandwidth each. Streams data over 100G Ethernet using AMD CMAC IP with flow control.
- **Teaching and supervision:** Taught the System-on-Chip Laboratory course and supervised 14 bachelor's and master's theses.

Research Assistant

2020 – 2021

Karlsruhe Institute of Technology (KIT), Karlsruhe

- Integrated an SSD controller IP core into an FPGA- and CPU-based data acquisition system for the Tandem-L radar satellite mission.

Working Student in Application Engineering

2019 – 2020

Infineon Technologies AG, Munich

- Performed PSpice circuit simulations and evaluated interface concepts between IGBT gate drivers and power management ICs.

Intern in Hardware and Logic Engineering

2019

Airbus Defence and Space, Ottobrunn

- Designed the digital PID controller for a proof-of-concept atomic clock as part of the bachelor's thesis.

EDUCATION

Dr.-Ing. (Ph.D.) Electrical Engineering and Information Technology <i>Summa cum laude, Karlsruhe Institute of Technology (KIT), Karlsruhe</i> Thesis: Template-Based Deployment for Real-Time Deep Learning on Point Clouds	2022 – 2026
M.Sc. Electrical Engineering and Information Technology <i>Grade 1.1 (with distinction), Karlsruhe Institute of Technology (KIT), Karlsruhe</i>	2019 – 2022
B.Sc. Electrical Engineering and Information Technology <i>Grade 1.7 (good), Karlsruhe Institute of Technology (KIT), Karlsruhe</i>	2016 – 2019

SELECTED PUBLICATIONS

- M. Neu**, F. Baptist, T. Lobmaier, F. Papagno, T. Ferber, J. Becker, "Reconfigurable Computing Challenge: Real-Time Graph Neural Networks for Online Event Selection in Big Science," in *Proc. IEEE FCCM*, Atlanta, USA, 2026.
- I. Haide, **M. Neu**, Y. Unno et al., "Real-time graph neural networks on FPGAs for the Belle II electromagnetic calorimeter," submitted to JINST, 2026.
- M. Neu**, I. Haide, T. Justinger, T. Rädler, V. Dajaku, T. Ferber, J. Becker, "Real-Time Graph-based Point Cloud Networks on FPGAs via Stall-Free Deep Pipelining," in *Proc. IEEE SBCCI*, Manaus, Brazil, 2025.
- C. Karle, **M. Neu**, "Data Acquisition Device for Acquiring and Generating Signals for a MIMO Antenna Device," German Patent Application DE 10 2023 121 976 A1, 2025.
- M. Neu**, J. Becker, P. Dorwarth et al., "Real-Time Graph Building on FPGAs for Machine Learning Trigger Applications in Particle Physics," *Computing and Software for Big Science*, vol. 8, no. 8, 2024.
- M. Neu**, C. Karle, B. Nuss, P. Groeschl, J. Becker, "A Scalable and Cost-efficient Antenna Testbed using FPGA-Server Compound Structures for Prototyping 6G Applications," in *Proc. IEEE DCOSS-IoT*, Coral Bay, Cyprus, 2023.

TECHNICAL SKILLS

Hardware Design

SystemVerilog, Chisel, Vitis HLS, AMD Versal AI Engines, AMD RFSocCs, Vivado IP integration (GT transceivers, RF data converters, 100G CMAC)

Compilers & Machine Learning Deployment

MLIR, LLVM, CIRCT, onnx-mlir, QONNX, graph neural networks, point cloud networks

Programming

C++, Python, Scala, Tcl

Tools & Verification

AMD Vivado, AMD Vitis, QuestaSim, cocotb, CMake, Git, Linux

AWARDS & HONORS

Leadership Talent Academy Scholarship <i>Karl Schlecht Foundation</i>	2021
Deutschlandstipendium (Germany Scholarship) <i>Federal Ministry of Research, Technology and Space</i>	2021

LANGUAGES

German (native), English (fluent)